

Crossdem Abstract: Comparing Politicians' Rhetoric With Democracy Indicators

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1 INTRODUCTION

Politicians are some of the most influential people in human society: from the Greeks' demagogues to our Presidents and Prime Ministers, leaders have always tried to steer society. Shapiro and Page (1984) [4] showed how nearly all 20th century U.S. Presidents have tried to lead public opinion, with varying degrees of success.

We can differentiate between two types of political speeches: *internal speeches*, aimed at other institutional bodies, such as parliamentary speeches, and *external speeches*, aimed at the population, such as public speeches, interviews and social network posts. This last category is especially interesting since it affords politicians a direct and high-volume communication channel to their potential followers.

On the other hand, internal speeches are especially easy to access, being already aggregated in public databases, yet their technical and codified nature makes their interpretation harder for both human and automated agents.

2 PROJECT GOAL

I want to compare politicians' speeches among three countries, namely Italy, France and the United States, between 1945 and 2025, and then cross-compare the resulting analyses with democracy level indicators taken from the V-Dem dataset¹ and fine-tune an LLM to better interpret political speech and classify populism and hate speech.

The first step is building a corpus of politicians' speeches, one corpus for Italian speeches, one for French speeches and one for American speeches. To limit the scope, I decided to take only speeches from Italian Prime Ministers, U.S. Presidents, and either French Presidents when the President and the Prime Minister come from the same party, or French Prime Ministers when the President and Prime Minister are in opposition to each other. This is justified by the fact that when they are politically aligned, the President dominates. During cohabitation (opposing political camps), the Prime Minister gains real policy autonomy [3].

¹The V-Dem Dataset is a multidimensional and disaggregated dataset that reflects the complexity of the concept of democracy as a system of rule that goes beyond the simple presence of elections: <https://www.v-dem.net/data/the-v-dem-dataset/>.

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Speeches can be taken from different sources, such as official speeches, interviews, or social network posts, and can be in different media formats, such as textual, audio or video. For the latter two, a translating tool will be needed (such as a speech-to-text library).

The dataset will have a fixed, structured taxonomy inspired by ParlaMint. For example, the table for Italian government will have a *government* column whose values will be `degasper i2` for "De Gasperi II" Government, `forlani` for "Forlani" government and so on.

It is likely that for older politicians there are fewer speeches available, i.e., the dataset becomes more sparse as we go back in time. Moreover, speeches will be in three languages, namely Italian, French and English. For this reason, data augmentation techniques may be useful.

For speech to text processing, OpenAI's Whisper library is a Transformer sequence-to-sequence model trained on various speech processing tasks, including multilingual speech recognition, speech translation, spoken language identification, and voice activity detection.

The second step is using the corpus to perform linguistic analyses. The corpus will be pre-processed, with procedures such as tokenization, stemming and lemmatization. We can then compute word frequency and text complexity measures, as well as hate speech and populism identification.

This can be done with known libraries such as spaCy, or with transformers from the Hugging Face libraries.

Sawicki et al. 2023's survey [5] tells us that English is by far the most represented language in the field of natural language processing. Fortunately, French and Italian are still in the top eight, so finding libraries already capable of handling them should be easy enough.

The third step is cross-referencing the linguistic metrics with democratic indicators such as freedom of expression, freedom of association, transparency and fairness of law and other such metrics. All democracy metrics can be found in the V-Dem dataset.

I must use the *CD Country-Date* dataset (version 16, released in March 2026) since, to keep track of Italian governments, year-level granularity is not enough.

The dataset covers Italy between 1861 and 2025, France between 1789 and 2025, and the U.S. between 1789 and 2025.

Some high level democracy indices (each normalized to a scale between zero and one): `v2x_polyarchy`, the *electoral* democracy index, `v2x_libdem`, the *liberal* democracy index, `v2x_partipdem`, the *participatory* democracy index, `v2x_delibdem`, the *deliberative* democracy index focusing on process, respectful dialogue, and the common good, and `v2x_egalidem`, the *egalitarian* democracy index considering social status, inequalities, and resource distribution.

The fourth step is to fine-tune a pre-trained transformer model, to make it better understand political speeches in three different languages, and to train it in a classification task for hate speech and populism recognition.

Card et al. [1] fine-tuned a pre-trained RoBERTa model achieving 65% accuracy in classifying Congressional Record entries as being pro-immigration, anti-immigration, or neutral, while Li et al. [2] found that LS-LLaMA substantially outperforms LLMs ten times its size in text classification tasks and demonstrates consistent improvements compared to robust baselines like BERT-Large and RoBERTa-Large.

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